

MATH 200 - Elementary Linear Algebra

MATH 201 - Intermediate Linear Algebra

Section 4.4 Notes

Spanning Sets and Linear Independence

Devotional Thought

Linear Combinations of Vectors in a Vector Space

Recall that when we say *vector space*^{*} we mean a subspace of R^n (including R^n itself).

Definition A vector $\mathbf{v}$ in a vector space V is a **linear combination** of the vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ in V when $\mathbf{v}$ can be written in the form

$$\mathbf{v} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_n\mathbf{u}_n$$

where $c_1, c_2, \dots, c_n$ are scalars.

Example [Example 1(a)] The set of vectors in R^3 ,

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{(1, 3, 1), (0, 1, 2), (1, 0, -5)\},$$

$\mathbf{v}_1$ is a linear combination of $\mathbf{v}_2$ and $\mathbf{v}_3$ because

$$\mathbf{v}_1 = 3\mathbf{v}_2 + \mathbf{v}_3.$$

^{*}As a further reminder, the definition of a vector space is a lot more general and abstract; it will be the first thing we discuss when it is just MATH 201 students

Example [Example 2] If possible, write the vector $\mathbf{v} = (1, 1, 1)$ as a linear combination of vectors in the set

$$S = \{(1, 2, 3), (0, 1, 2), (-1, 0, 1)\}.$$

Example [Example 3] If possible, write the vector $\mathbf{w} = (1, -2, 2)$ as a linear combination of vectors in the set

$$S = \{(1, 2, 3), (0, 1, 2), (-1, 0, 1)\}.$$

Spanning Sets

If every vector in a vector space can be written as a linear combination of vectors in a set S , then S is a **spanning set** of the vector space.

Definition Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be a subset of a vector space V . The set S is a **spanning set** of V when every vector in V can be written as a linear combination of vectors in S . In such cases, it is said that S **spans** V .

Examples [Examples 4(a) and 6]

(a) The set $S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ spans R^3 because any vector $\mathbf{u}$ in R^3 can be written as the linear combination

$$\mathbf{u} = (u_1, u_2, u_3) = u_1(1, 0, 0) + u_2(0, 1, 0) + u_3(0, 0, 1).$$

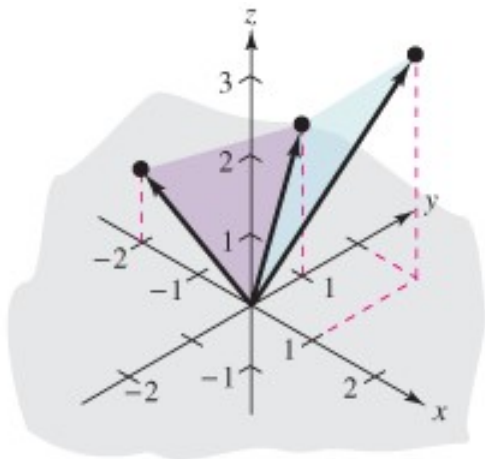
This set S is called the **standard spanning set** of R^3 .

(b) From Examples 2 and 3 we know that the set

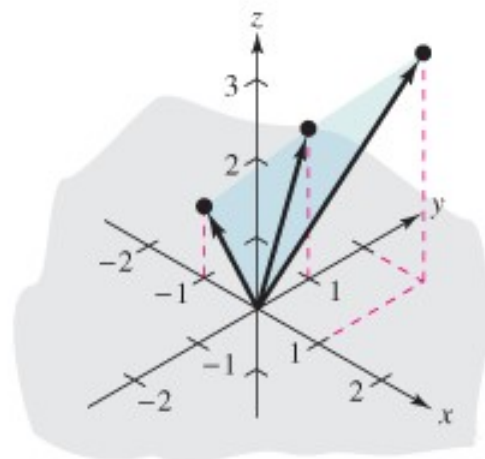
$$S = \{(1, 2, 3), (0, 1, 2), (-1, 0, 1)\}$$

does not span R^3 ; even though $\mathbf{v} = (1, 1, 1)$ *can* be expressed as a linear combination of vectors in S , the vector $\mathbf{w} = (1, -2, 2)$ *cannot* be written as such. Thus S *does not span R^3* since *not every* vector in R^3 can be written as a linear combination of vectors in S .

Example [Example 5] Show that the set $S = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\}$ spans $\mathbb{R}^3$.



$S_1 = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\}$
 The vectors in S_1 do not lie
 in a common plane.



$S_2 = \{(1, 2, 3), (0, 1, 2), (-1, 0, 1)\}$
 The vectors in S_2 lie in a
 common plane.

Comparing the sets of vectors in the last two examples, note that the sets are the same except for a seemingly insignificant difference in the third vector.

$$S_1 = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\}$$

$$S_2 = \{(1, 2, 3), (0, 1, 2), (-1, 0, 1)\}$$

The difference, however, is significant, because the set S_1 spans R^3 whereas the set S_2 does not. The reason for this difference can be seen in the figure as the vectors in S_2 lie in a common plane; the vectors in S_1 do not.

Although the set S_2 does not span all of R^3 , it does span a subspace of R^3 —namely, the plane in which the three vectors of S_2 lie. This subspace is called the **span of S_2** .

Definition If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a set of vectors in a vector space V , then the **span of S** is the set of all linear combinations of the vectors in S ,

$$\text{span}(S) = \{c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k : c_1, c_2, \dots, c_k \text{ are real numbers}\}.$$

We may also denote this set as $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$.

For a nonempty subset S of a vector space V , the following statements are equivalent.

- S is a spanning set of V .
- V is spanned by S .
- S spans V .
- $\text{span}(S) = V$.

Based on the last statement, it should not be surprising that $\text{span}(S)$ is always a subspace of V .

Theorem 4.7 If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a set of vectors in a vector space V , then $\text{span}(S)$ is a subspace of V . Moreover, $\text{span}(S)$ is the smallest subspace of V that contains S , in the sense that every other subspace of V that contains S must contain $\text{span}(S)$.

Remark It is important to remember that *a set of vectors S will always span a vector space*. For every subset S of vectors in a vector space V , $\text{span}(S)$ is a subspace of V , meaning it *is* a vector space. *The issue at hand is whether S spans the whole space V or just some proper subspace W of V .*

Examples [Exercises 10, 14, 15, and 18] Determine whether the set S spans R^2 . If the set does not span R^2 , then give a geometric description of the subspace that it does span.

- $S = \{(-1, 1), (3, 1)\}$

- $S = \{(1, 1)\}$

- $S = \{(-1, 2), (2, -4)\}$

- $S = \{(-1, 2), (2, -1), (1, 1)\}$

Example [Exercise 22] Determine whether the set S spans R^3 . If the set does not span R^3 , then give a geometric description of the subspace that it does span.

$$S = \{(1, 0, 1), (1, 1, 0), (0, 1, 1)\}$$

One main takeaway about spanning is that it gives the *existence* of a linear combination representation for elements in a vector space.

Question Is there *uniqueness* of such a linear combination representation?

Linear Dependence and Linear Independence

Recall that for a set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ in a vector space V , the vector equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k = \mathbf{0}$$

always has the trivial solution $c_1 = 0, c_2 = 0, \dots, c_k = 0$.

Sometimes, however, there are also nontrivial solutions. For instance, in the first example of this section we saw that in the set

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{(1, 3, 1), (0, 1, 2), (1, 0, -5)\}$$

the vector $\mathbf{v}_1$ can be written as a linear combination of the other two vectors,

$$\mathbf{v}_1 = 3\mathbf{v}_2 + \mathbf{v}_3.$$

So, the vector equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{0}$$

has a nontrivial solution in which the coefficients are *not all zero*:

$$c_1 = 1, \quad c_2 = -3, \quad c_3 = -1.$$

When a nontrivial solution exists, the set is **linearly dependent**. Had the only solution been the trivial one ($c_1 = c_2 = c_3 = 0$), then the set S would have been **linearly independent**. This concept is essential to the study of linear algebra as it provides *uniqueness* to linear combination solutions.

Definition A set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ in a vector space V is **linearly independent** when the vector equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$$

has only the trivial solution

$$c_1 = 0, \ c_2 = 0, \ \dots, \ c_k = 0.$$

If there are also nontrivial solutions, then S is **linearly dependent**.

Examples [Example 7] The following subsets of R^2 are all linearly dependent.

(a) $S = \{(1, 2), (2, 4)\}$

(b) $S = \{(1, 0), (0, 1), (-2, 5)\}$

(c) $S = \{(0, 0), (1, 2)\}$

Remark Any set of vectors containing $\mathbf{0}$ will be linearly dependent.

Example [Example 8] Determine whether the set of vectors in R^3 is linearly independent or linearly dependent.

$$S = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\}$$

Testing for Linear Independence and Linear Dependence

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be a set of vectors in a vector space V . To determine whether S is linearly independent or linearly dependent, use the steps below.

1. From the vector equation $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$, write a system of linear equations in the variables $c_1, c_2, \dots, c_k$.
2. Determine whether the system has a unique solution.
3. If the system has only the trivial solution, $c_1 = 0, c_2 = 0, \dots, c_k = 0$, then the set S is linearly independent. If the system also has nontrivial solutions, the S is linearly dependent.

Example [Exercise 32] Determine whether the set S is linearly independent or linearly dependent.

$$S = \{(6, 2, 1), (-1, 3, 2)\}$$

As a consequence of the F.T.E.L.A., a subset S of R^n consisting of n vectors is linearly independent if and only if the determinant of the coefficient matrix for the corresponding homogeneous system is zero.

Example [Example 8] Determine whether the set of columnn vectors in R^4 is linearly independent or linearly dependent.

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \\ 2 \end{bmatrix} \right\}$$

Remark Because of the homogeneous systems involved, determining linear independence is inherently easier than deducing whether a set spans a vector space.

If a set of vectors is linearly dependent, then by definition the equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$$

has a nontrivial solution (a solution for which not all the c_i s are zero). For instance, if $c_1 \neq 0$, then you can solve this equation for $\mathbf{v}_1$ and write $\mathbf{v}_1$ as a linear combination of the other vectors $\mathbf{v}_2, \mathbf{v}_3, \dots$, and $\mathbf{v}_k$. In other words, the vector $\mathbf{v}_1$ *depends* on the other vectors in the set. This property is characteristic of a linearly dependent set.

Theorem 4.8 A set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$, $k \geq 2$ is linearly dependent if and only if at least one of the vectors $\mathbf{v}_i$ can be written as a linear combination of the other vectors in S .

The proof of this fact is actually quite computational. (See the last slide of this section.)

There is also a more simplistic way to think about Theorem 4.8 for just two vectors.

Corollary Two vectors $\mathbf{u}$ and $\mathbf{v}$ in a vector space V are linearly dependent if and only if one is a scalar multiple of the other.

Example [Exercise 55] Show that the set is linearly independent by finding a nontrivial linear combination of vectors in the set whose sum is the zero vector. Then express one of the vectors in the set as a linear combination of the other vectors in the set.

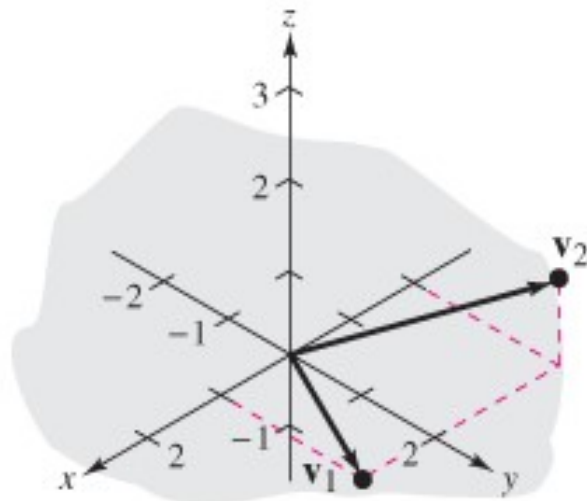
$$S = \{(1, 1, 1), (1, 1, 0), (0, 1, 1), (0, 0, 1)\}$$

Examples [Example 13]

(a) The set $S = \{(1, 2, 0), (-2, 2, 1)\}$ is linearly independent.

(b) $S = \{(4, -4, -2), (-2, 2, 1)\}$ is linearly dependent.

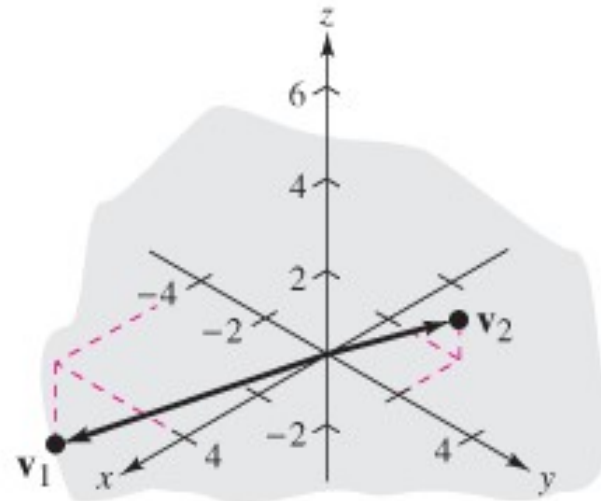
a.



$$S = \{(1, 2, 0), (-2, 2, 1)\}$$

The set S is linearly independent.

b.



$$S = \{(4, -4, -2), (-2, 2, 1)\}$$

The set S is linearly dependent.

Proof of Theorem 4.8.

($\Rightarrow$) Assume S is a linearly dependent set. Then there exist scalars $c_1, c_2, c_3, \dots, c_k$ (not all zero) such that

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}.$$

One of the coefficients must be nonzero; without loss of generality assume $c_1 \neq 0$. Then solving for $\mathbf{v}_1$ as a linear combination of the other vectors produces

$$c_1 \mathbf{v}_1 = -c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 - \cdots - c_k \mathbf{v}_k$$

$$\mathbf{v}_1 = -\frac{c_2}{c_1} \mathbf{v}_2 - \frac{c_3}{c_1} \mathbf{v}_3 - \cdots - \frac{c_k}{c_1} \mathbf{v}_k$$

So $\mathbf{v}_1$ is a linear combination of $\mathbf{v}_2, \mathbf{v}_3, \dots$, and $\mathbf{v}_k$.

($\Leftarrow$) Conversely, assume one of the vectors in S is a linear combination of the other vectors. Without loss of generality suppose $\mathbf{v}_1$ is such a vector. That is,

$$\mathbf{v}_1 = c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 + \cdots + c_k \mathbf{v}_k$$

Then the equation $-\mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 + \cdots + c_k \mathbf{v}_k = \mathbf{0}$ has at least one coefficient, -1 , that is nonzero, and we can conclude that S is linearly dependent. ■

More Exercises (for extra practice as time allows)

Example [Exercise 58] For which values of t is each set linearly independent?

(a) $S = \{(t, 0, 0), (0, 1, 0), (0, 0, 1)\}$

(b) $S = \{(t, t, t), (t, 1, 0), (t, 0, 1)\}$